

Santiago Segovia

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EDUCATION

John Hopkins University

Master of Science in Data Science

Aug. 2026 - Present

University of Houston

Bachelor of Science in Computer Science, Bachelor of Science in Mathematics

Aug. 2022 - Dec. 2025

Cum Laude - GPA: 3.732

EXPERIENCE

Math Teacher

Harmony Virtual Academy

Aug. 2026 – Present

Houston, TX

- * Instruct **100+** students across on-level and Pre-AP Algebra II, delivering coursework centered on mathematical reasoning, problem-solving, and quantitative analysis
- * Lead an **after-school Coding Club for 10+ students**, teaching Python, Git, version control, and foundational programming concepts through hands-on projects

Teacher Assistant - Automata and Computability

University of Houston - Computer Science Department

Jan. 2025 – Dec. 2025

Houston, TX

- * Mentored **100+** students in automata theory, formal languages, and computability through office hours and exam review sessions
- * Graded homework, quizzes, and exams while providing constructive feedback

PROJECTS

SSBU Self-Performance Analytics | *Python, Pandas, Scikit-learn, XGBoost, SHAP*

Aug. 2026 - Sep. 2026

- * Built an end-to-end ML pipeline on **150+ self-collected matches**, engineering gameplay and matchup features and applying K-Means, silhouette analysis, and PCA to identify distinct playstyle and matchup-performance profiles
- * Trained an **XGBoost classifier** achieving **0.84 ROC-AUC** on holdout data; used SHAP to identify damage efficiency as a key predictive signal and found controlled matches yielded a **13 percentage-point higher win rate** than high-pressure matches

Portfolio Website | *Next.js, React, TypeScript, Tailwind CSS, Framer Motion*

May. 2026 - Present

- * Developed and deployed a responsive portfolio using Next.js, React, and TypeScript featuring animated mathematical visualizations and secure contact functionality
- * Deployed on Vercel with Resend API integration and server-side validation

Sparse Matrix Optimization for Homomorphic Inference | *MATLAB*

Aug. 2025 – Dec. 2025

- * Developed a MATLAB encrypted-inference framework evaluating **random, block, banded, and row-column sparsity**, tracking encrypted operations, runtime, and classification accuracy
- * Analyzed **cost-accuracy Pareto tradeoffs** across tabular and MNIST datasets, identifying structured sparsity configurations that reduced encrypted computation while preserving predictive performance

GPU-Accelerated LU Solver | *CUDA, C++, cuBLAS*

Jan. 2025 - May. 2025

- * Developed a CUDA/C++ LU decomposition solver for dense matrices, achieving **41× speedup** over a CPU implementation using shared-memory optimization and cuBLAS
- * Optimized GPU memory access patterns through kernel profiling, improving throughput and reducing execution overhead

Amazon Review Opinion Search Engine | *Python*

Jun. 2025 – Aug. 2025

- * Built an opinion-based product review search engine using multiple retrieval models and evaluated ranking quality with Precision@k
- * Improved search relevance through iterative query refinement and retrieval model evaluation

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, MATLAB, PHP, R

Frameworks & Libraries: React, Next.js, Node.js, PyTorch, pandas, NumPy, scikit-learn, XGBoost, SHAP

Databases: MySQL, PostgreSQL

Tools: Git, CUDA, cuBLAS, VS Code